Lesson plan · for the teacher

# Data all around the school

Data engineering · Level 1 · Lesson 1

**Time**
40 min

**Materials**
Board and chalk; A number card for each pupil (1, 2, 3…); Optional: a phone or computer with a spreadsheet app

## Learning goals

- Say what data is, and find it in our school.
- Read a table: rows, columns and cells.
- Build a tidy table of our class, and count from it.

## Activity: Our class in a table

1. Warm-up (5 min): Ask pupils what the school writes down every day: the register, marks, the rain gauge, library cards. List their ideas on the board and circle the ones kept as tables.
2. Teach (8 min): Draw a small register on the board. Point to a row (one pupil), a column (one day) and a cell (one letter, P or A). Then show a messy version, with a Total row and a cell holding two values, and ask what goes wrong when we count.
3. Build (15 min): Draw a table with the columns “Pupil no.”, “How I come to school” and “Minutes”. Agree the codes first (W walk, B bus, C bicycle, M motorbike). Each pupil comes up and fills in their row, using their number, not their name.
4. Count (7 min): Count each code with tally marks beside the table, then count the rows with more than 30 minutes. Check that the counts add up to the number of pupils present.
5. Wrap-up (5 min): Ask which rules made the table easy to count: one row per pupil, one fact per column, one value per cell. Then give the exit ticket.

**Give each pupil a number before the lesson, and keep the list of numbers and names to yourself. If some pupils don’t know their minutes, accept “about” values and say so on the table.**

## Discussion

- Which data does our school keep that nobody ever counts?
- Why did we use numbers instead of names in our table?
- What went wrong when the table had a Total row in it?

## Exit ticket

- Draw a tiny table with two rows and two columns about anything you like, and label a row, a column and a cell.

**Try it in class tomorrow**
Draw a table on the board: one row for each pupil, by number, not by name, and columns for how they come to school and the minutes it takes. Then count together: how many walk more than thirty minutes?

Worksheet · for pupils

# Data all around the school

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Name _____ Class _____ Date _____

**1** **A table of how pupils come to school and how long it takes: what should one row be?**

◯ One pupil ◯ One way of travelling ◯ The whole class

**2** **Match each value from a class table to its kind.**

| | | |
|---|---|---|
| 25 (minutes to walk to school) | • • • • | Text: read by people |
| Walks / bus / bicycle | • • • • | A date: written one agreed way |
| 2 March 2026 | • • • • | A category: it can be counted |
| “Always helps others” | • • • • | A number: it can be added and averaged |

**3** **A Total row inside the data helps a computer add up the class.**

◯ True ◯ False

**4** **Which of these break the rules of a tidy table? Choose all that apply.**

☐ The class written once, with the cells below left blank ☐ A Total row between the classes

☐ A column headed “Minutes to school” ☐ “14 / 17” in one cell ☐ One row for each pupil

✎ **Which data does your school already keep that could answer a real question, if it were in one tidy table?**